

On the terminal run the following command and note down the port of Galileo (in our case it is ACM0):

```
ls /dev/ttyACM*
```

In case you see an error message Java not found, install Java by running the following command:

```
sudo apt-get install default-jre
```

To select Serial Port, go to Tools→Serial Port→/dev/ttyACM0 (Fig. 11).

Software installation on Mac OS

For Mac, unzip the application and save it to the application folder. If

you have other Arduino versions, you can save the file by any name, only keep in mind that there should be no spaces in the name. Run the application by double-clicking on it (Fig. 13).

Driver installation. Once software installation is complete, you need to install the driver using the following steps (also shown in Fig. 14):

Connect 5V DC power to the board and then the USB cable (power should always be the first connection on Galileo board).

Allow a few seconds for the board to boot up. Open System Information window to check if the board has been detected properly. Path for the same is: Hold Options→Apple menu→System Information. Under USB, check Gadget Serial v2.4.

Under Network, check the name of your Galileo device; it should be similar to usbmodemxxxx. In this case it is fa21. Note the number as you will need it while uploading the code.

Go to Tools→Serial Port→/dev/

cu.usbmodemfa21 as shown in Fig. 15.

Uploading the code

The procedure for uploading Blink example code is the same for Windows, Linux and Mac OSes. To upload Blink example sketch, select the code from the path File→Examples→01.Basics→Blink (Fig. 16).

A new code will open. Click on Upload to upload the code (sketch) on the board. Status of the uploading is displayed on the bottom of the screen (Fig. 17).

When the upload is complete, you will see Transfer Complete message and pin 13 LED blinking on Intel Galileo board.

Some more examples

After successfully running our first LED blinking example, we can proceed to more examples like making LEDs to blink at various speeds, driving multiple LEDs with sequential effects, reading analogue input using simple potentiometer and driving RGB LEDs. So let us begin!

You need a breadboard, resistor, an LED and some jumper wires to set up the hardware. Though the board already has an LED connected to pin 13, we will connect an external LED to feel the experiment. Fritzing software comes handy when you do not have actual components with you. Here, we will use Intel Galileo Gen2 board image in Fritzing environment. (Intel Galileo Gen2 is the upgraded version of Intel Galileo.)

Blinking LED.

Wire up the circuit with a 330-ohm resistor using Fritzing as shown in Fig. 18. You can change the blinking speed by varying the delays in blink.ino program.

Multiple LEDs.

Let us move one step further and connect multiple LEDs. Connect the hardware

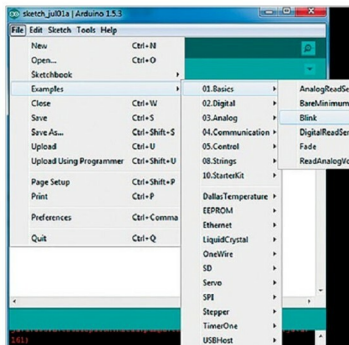


Fig. 16: Uploading Blink example code on Windows



Fig. 17: Status of uploading the code

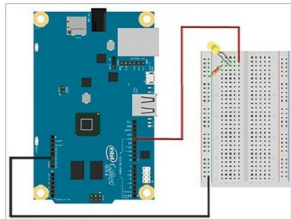


Fig. 18: Circuit for LED blinking example

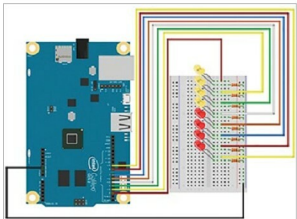


Fig. 19: Circuit for multiple blinking LEDs